IMPERIAL

Deep Learning

Lecture 8: Design an Agent

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What Have We Learned?

Core Topics:

- Neural networks and deep learning
- Convolutional neural networks
- Markov chains
- Markov Decision Processes
- Reinforcement learning
- Transformers and large language models

From Models to Agents

An AI agent is a system that:

1. Observes its environment
2. Builds an internal representation of the world
3. Takes actions to achieve a goal

Agent Architecture

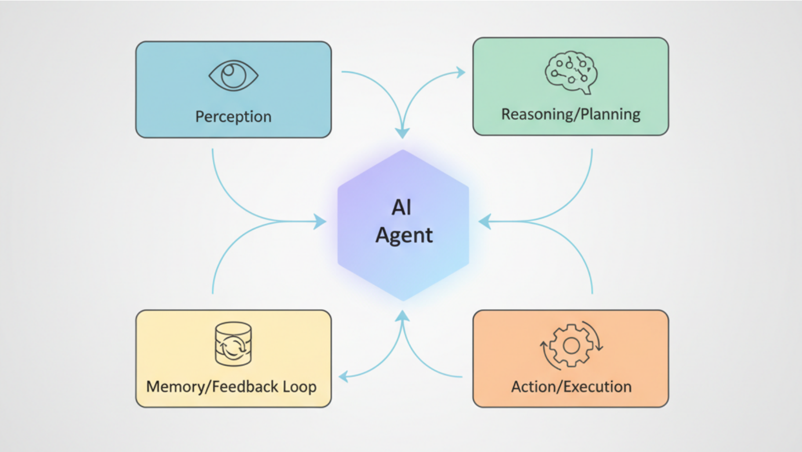

Figure: Source: Medium, Sudarshan Poudel

Today's Activity

Today you will design an **AI agent** to solve a task.

You will work in **small groups**.

Your goal is not only to describe the idea, but to **formulate the whole learning pipeline**, from where I collect the data to how I train each component.

Important things to consider

- Where do I find training data?
- Can I share my data?
- How does this system make decisions?
- Is it safe?
- What happens if it interacts with other agents? And what about other agents and people?

For Your Assigned Problem, Define the Following

For your agent, your group should define:

1. the **state space** S,
2. the **action space** A,
3. the **reward function** R,
4. the **learning method**,
5. the **sensors**,
6. the **memory**,
7. the **hardest technical challenge**.

Be as concrete as possible.

Do not just describe the application in a hand-wavy: formulate the whole architecture with as much detail as possible.

Autonomous Taxi Agent

Design an AI agent that controls an autonomous taxi in a city.

The agent must:

- navigate through urban traffic
- obey road rules
- avoid collisions
- deliver passengers efficiently

Possible observations:

- vehicle position and velocity
- nearby vehicles and pedestrians
- traffic lights and road signs
- map information

Autonomous Taxi Agent: Additional Complication

During heavy rain, several sensors become unreliable.

Examples:

- cameras may fail to detect lane markings
- pedestrians may appear suddenly
- GPS may become inaccurate in dense areas

How should the agent behave when its observations are uncertain?

Aircraft Co-Pilot Agent

Design an AI agent that assists a human pilot during a flight.

The system may help with:

- monitoring instruments
- detecting anomalies
- suggesting control actions
- assisting during difficult flight conditions

Possible observations:

- altitude, speed, heading
- engine status
- weather conditions
- pilot inputs

Aircraft Co-Pilot Agent: Additional Complication

During a storm, one of the aircraft sensors starts reporting incorrect data.

The agent receives conflicting information from different instruments.

- altitude readings disagree
- airspeed measurements fluctuate
- weather conditions are deteriorating

How should the system reason under conflicting observations?

Autonomous Cargo Ship Agent

Design an AI agent that controls a cargo ship travelling between ports.

The system must:

- follow a planned route
- avoid collisions with other vessels
- adapt to weather conditions
- optimise travel time and fuel usage

Possible observations:

- GPS position
- radar signals
- nearby vessel information
- weather forecasts

Autonomous Cargo Ship Agent: Additional Complication

In narrow shipping channels, many ships are present simultaneously.

Other vessels may behave unpredictably.

- some vessels do not follow expected navigation rules
- radar signals may be noisy
- congestion creates delayed consequences for actions

How should the agent plan under these conditions?

Autonomous Drone Delivery Agent

Design an AI agent that controls a drone delivering packages in an urban environment.

The agent must:

- navigate to delivery locations avoiding buildings, trees, and other drones
- adapt to changing weather conditions
- manage limited battery life

Possible observations:

- GPS position and velocity, IMU
- onboard camera or lidar data
- battery level
- map information

Autonomous Drone Delivery Agent: Additional Complication

During part of the route, the drone loses reliable GPS signal.

At the same time:

- wind conditions become unstable
- battery is running low
- another drone is approaching the same delivery area

How should the agent behave when localisation and planning become uncertain?

Final Question

After designing your system, think about what is the hardest technical challenge?

Examples:

- exploration
- safety
- partial observability
- reward design
- data limitations

Closing Thought

What happens when millions of AI agents interact?

Examples:

- financial markets
- traffic systems
- energy grids
- online platforms

Understanding the emergent dynamics of these systems is an open research problem.

Any questions?